

Exploratory Data Analysis (EDA) Report

1. Introduction

Objective:

The objective of this analysis is to explore the sales dataset, identify patterns, understand customer behaviour, detect data quality issues, and derive meaningful business insights using Exploratory Data Analysis (EDA).

2. Dataset Overview

Dataset Details

- Number of Records: **1000**
- Number of Features: **12**
- Dataset Type: Sales Data

Features Included

- Order ID
- Order Date
- Customer ID
- Customer Name
- Age
- Gender
- City
- Product
- Category
- Quantity
- Unit Price
- Total Sales

3. Data Cleaning

The following preprocessing steps were performed:

- Missing values in **Age** were filled using the median.

- Missing values in **City** were filled using the mode.
- Order Date was converted into datetime format.
- Duplicate Order IDs were removed for SQL analysis.
- Data types were verified and corrected where necessary.

4. Descriptive Statistics

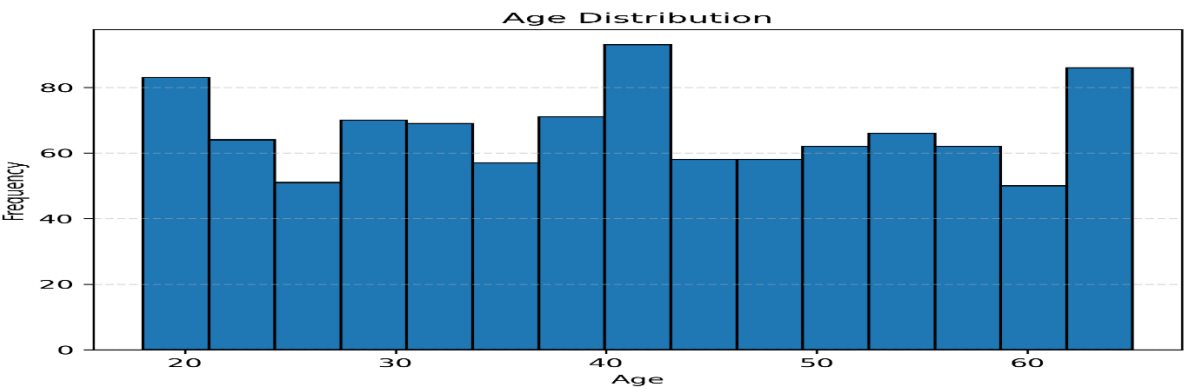
Statistic	Age	Quantity	Unit Price	Total Sales
Count	1000	1000	1000	1000
Mean	41.35	5.43	25,486.78	139,399.44
Min	18	1	145.78	437.34
25%	30	3	13,895.72	47,066.63
50%	41	5	25,398.74	108,594.03
75%	53	8	37,512.38	203,722.88
Max	65	10	49,997.53	493,677.50
Std Dev	13.68	2.84	14,179.40	114,100.05

Key Findings

- Average customer age is approximately **41 years**.
- Average quantity purchased is approximately **5 units**.
- Average unit price is around **₹25,487**.
- Average total sales per order are approximately **₹139,399**.

5. Univariate Analysis

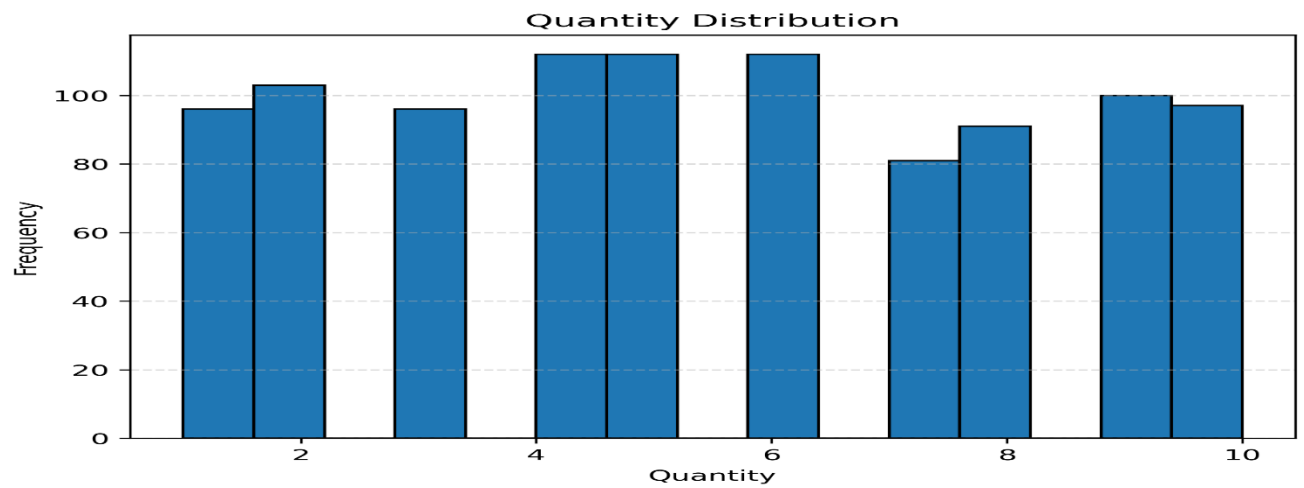
Age Distribution



Insight

Customer ages are fairly distributed between 18 and 65 years.

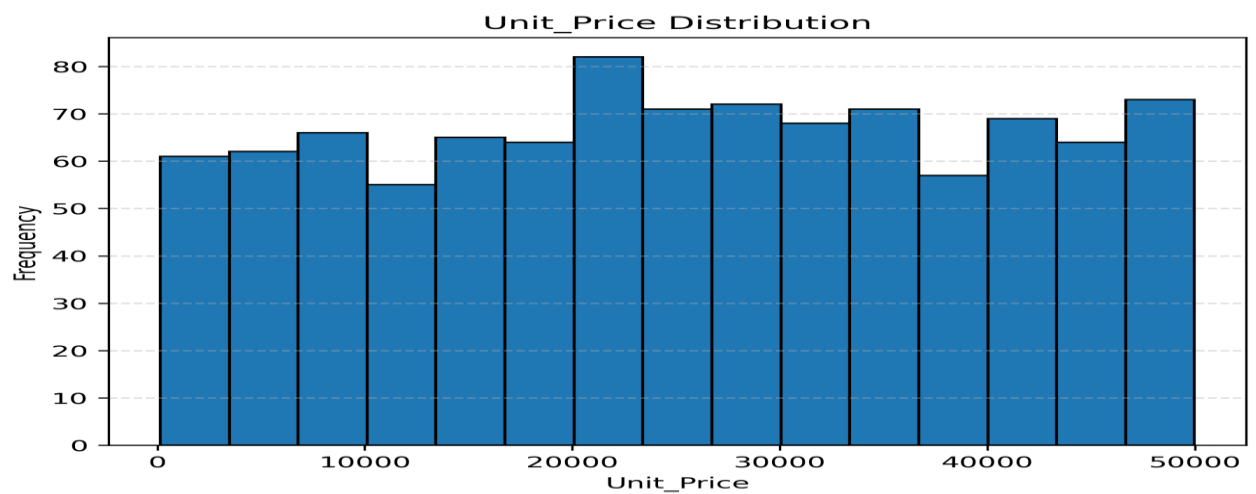
Quantity Distribution



Insight

Most orders contain between 3 and 8 products.

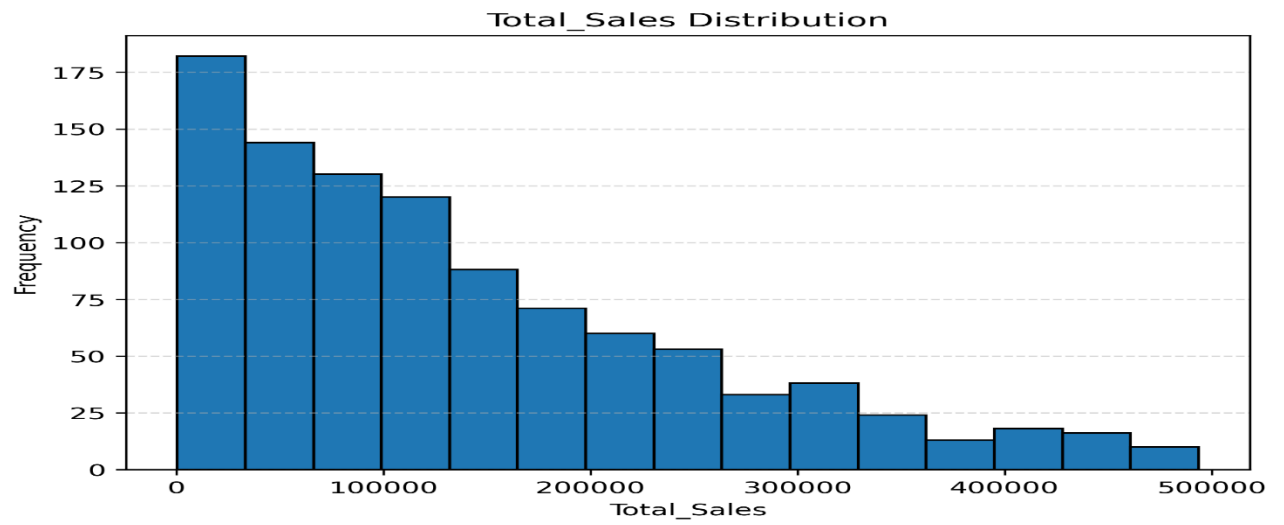
Unit Price Distribution



Insight

Products are distributed across a wide price range.

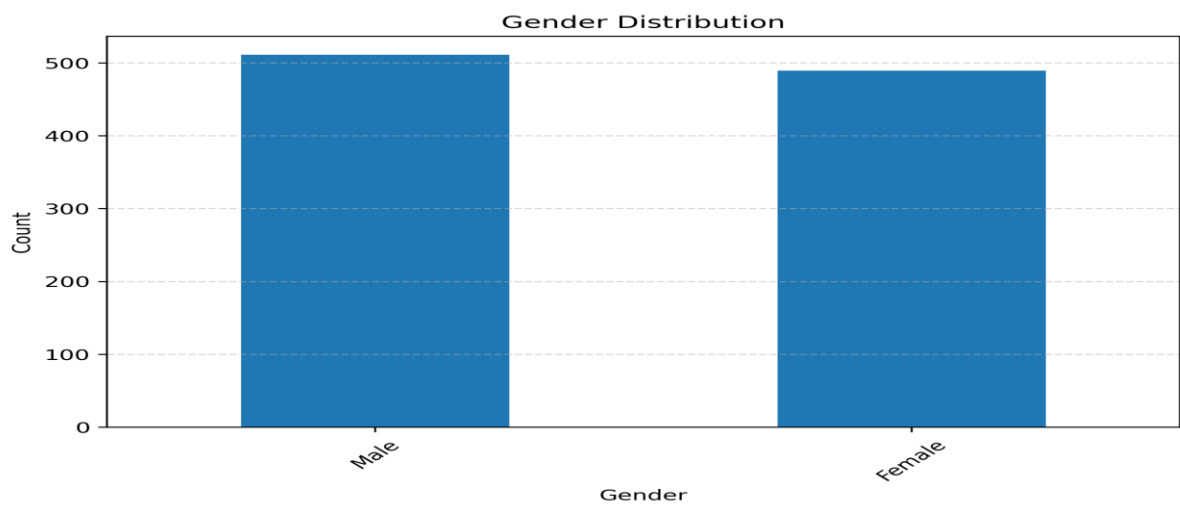
Total Sales Distribution



Insight

Most sales fall in the lower-to-middle range, with a few high-value orders.

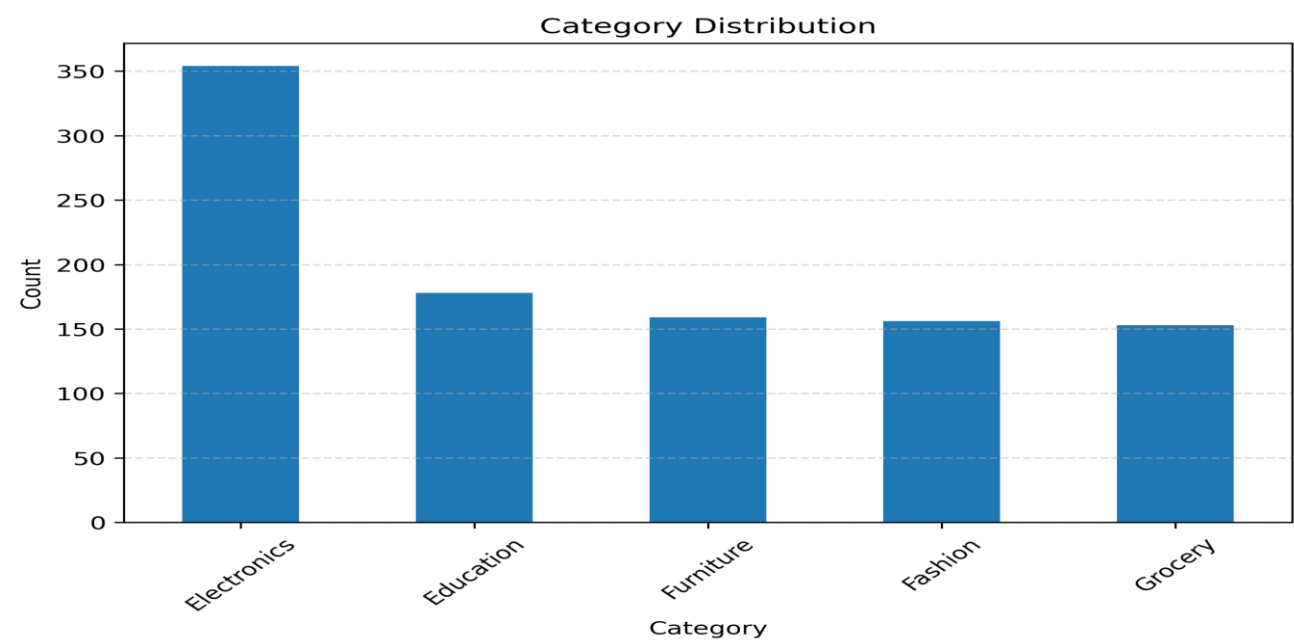
Gender Distribution



Insight

Male and female customers are almost equally represented.

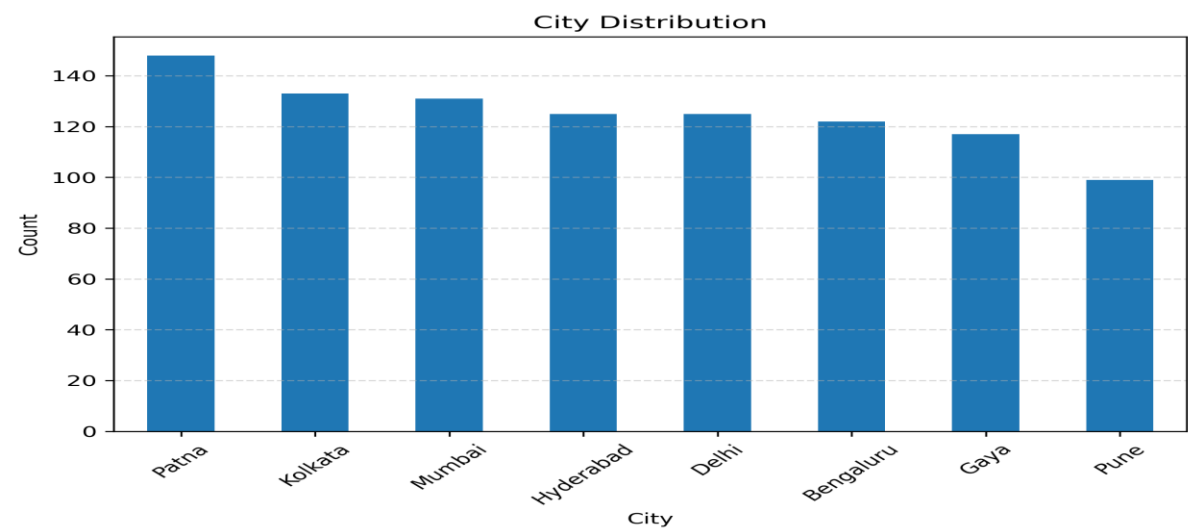
Category Distribution



Insight

The dataset contains five product categories with a balanced distribution.

City Distribution

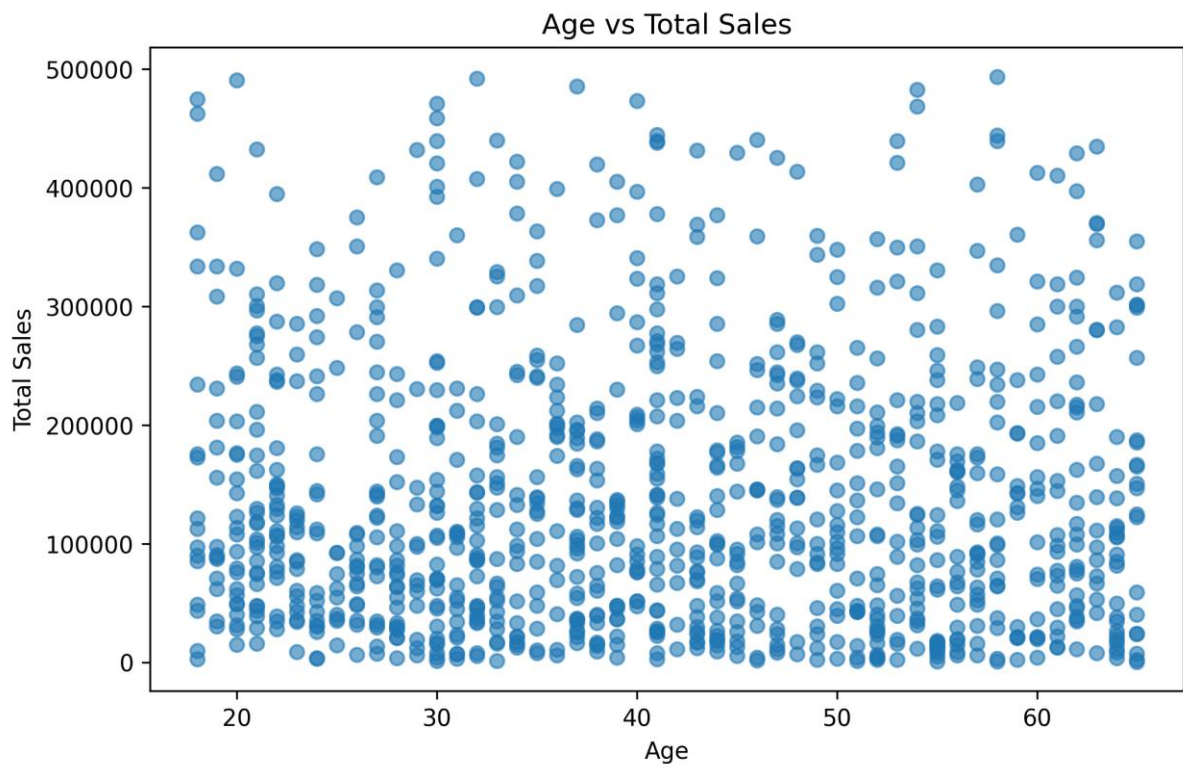


Insight

Sales records are distributed across multiple cities.

6. Multivariate Analysis

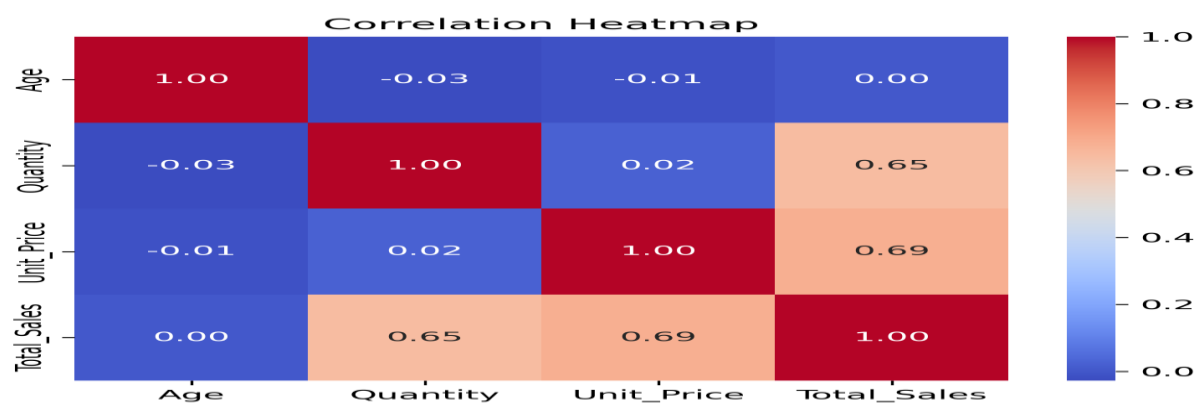
Scatter Plot (Age vs Total Sales)



Insight

No strong relationship exists between customer age and total sales.

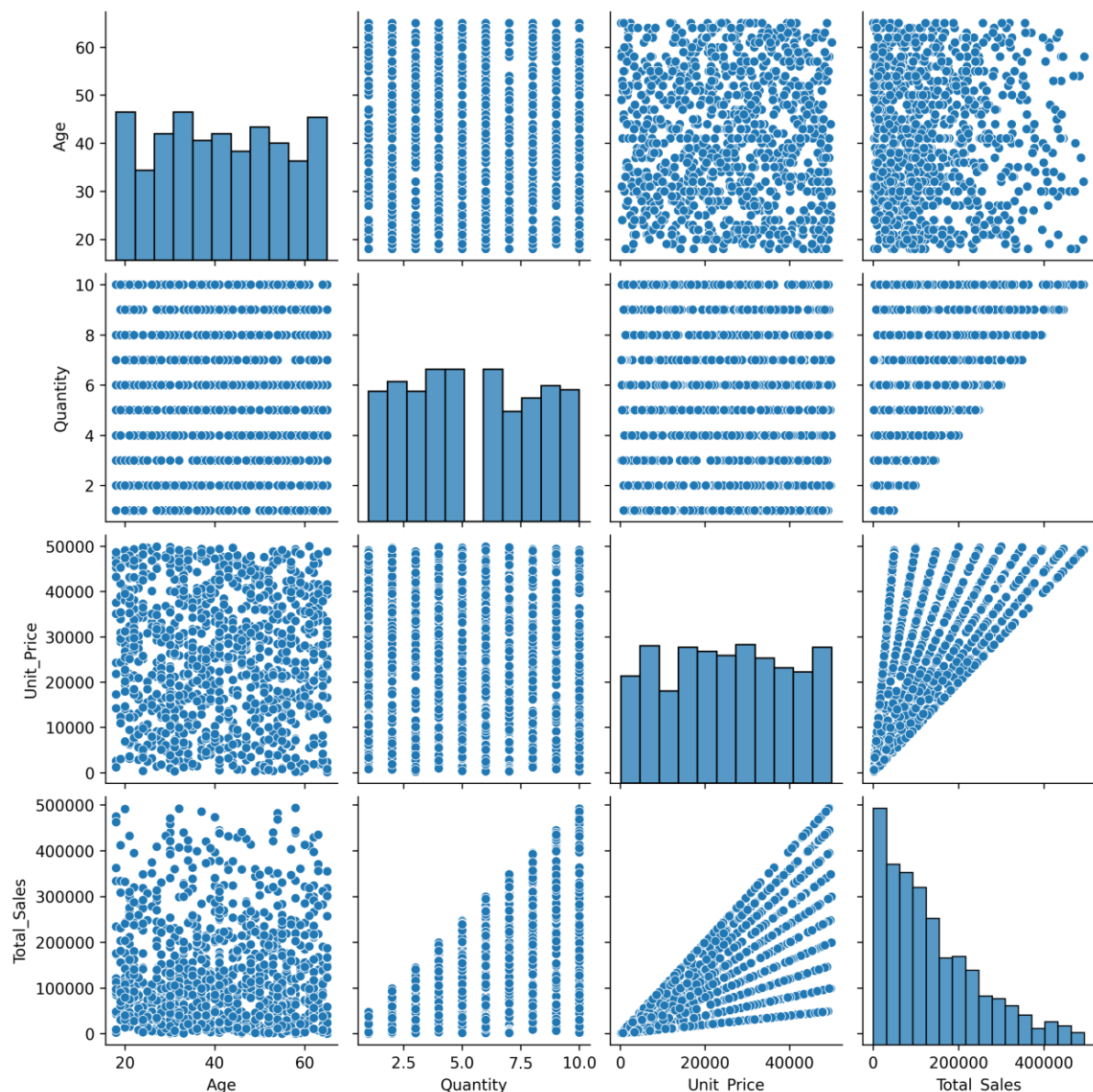
Correlation Heatmap



Insight

- Unit Price and Total Sales show a moderate positive correlation.
- Quantity also positively influences Total Sales.
- Age has little correlation with sales-related variables.

Pair Plot



Insight

The pair plot helps visualize relationships and detect potential outliers among numerical variables.

7. Key Findings

- Customers aged **36–45** contribute the highest revenue.
- **Electronics** is the highest-performing category.
- **Laptop** is the best-selling product.
- **Patna** generates the highest sales.
- **Unit Price** has the strongest influence on **Total Sales**.

8. Conclusion

The EDA provided valuable insights into **customer behaviour, sales trends**, and **product performance**. The findings can support better inventory planning, targeted marketing, and strategic business decision-making.